

Week 09: Specifications and induction

Functional Programming in Lean

Learning goals

- ▶ Read a theorem as a specification
- ▶ Follow the base and induction cases
- ▶ Audit which axioms a proof uses

Central idea

A recursive function suggests the induction structure for a property about it.

Lean example

```
theorem append_nil (xs : List  $\alpha$ ) :  
  append xs [] = xs := by  
  induction xs with  
  | nil => rfl  
  | cons x xs ih => simp [append, ih]
```

What to notice

- ▶ The induction hypothesis speaks about the tail
- ▶ `simp` uses definitions and known equalities
- ▶ `#print axioms` exposes proof dependencies

Live coding

1. Read the goal after induction `xs`
2. Complete the cons case
3. Explain the final theorem in English

Pause and predict

What exactly does `ih` state in the `cons` case?

Exercise connection

Open `exercises/Week09.md` after the lecture.

The complete example is `Course/Week09.lean`; the fill-in version is `Templates/Week09.lean`.